



Networking & Cloud Fundamentals

This overview brings together foundational material on cloud services, core IP-address concepts, IPv4 subnetting methods, and basic Cisco switching and routing configuration.



Content Sources

- Section 1: Overview of **cloud computing** and **AWS** fundamentals from a 28-page PDF document. ([source](#))
- Section 2: Concise reference on **IP addressing** and **subnet masks** from a 14-page PDF document. ([source](#))
- Section 3: Instruction on **IPv4 subnetting** techniques for **network segmentation** from a 44-page PDF document. ([source](#))
- Section 4: Essentials of **Cisco switching** and **device configuration** from a 64-page PDF document. ([source](#))

Section 1: Cloud Computing Basics & AWS Overview

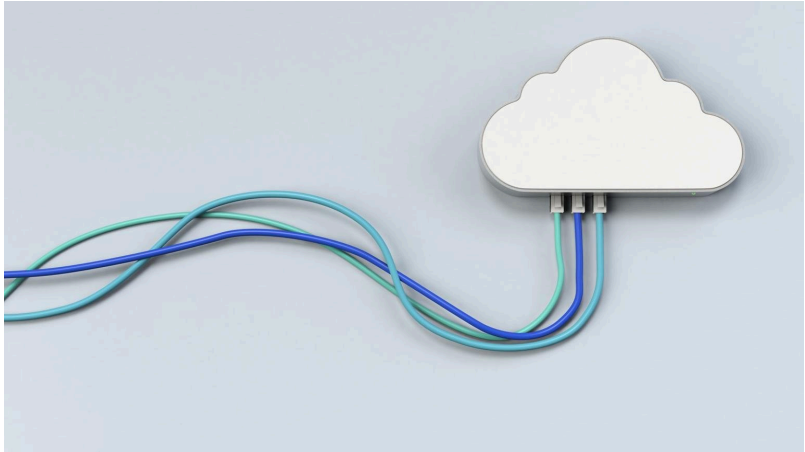
This section covers cloud computing fundamentals and Amazon Web Services from a 28-page PDF document ([source](#))

Cloud Computing Fundamentals

Cloud computing delivers technology services (storage, databases, compute power) via the Internet rather than maintaining on-premise hardware.

Key characteristics include no hardware ownership, remote data storage, pay-as-you-go pricing, and elastic scalability

Benefits: Cost-effectiveness, operational efficiency, and flexibility



Cloud Deployment Models

Model	Core Traits	Typical Use-Cases
Public Cloud	Services over Internet; shared resources	Cost-sensitive workloads, rapid prototyping
Private Cloud	Single-organization dedicated	Sensitive data, compliance environments
Hybrid Cloud	Public + private combination	Flexibility, optimized performance

Amazon Web Services (AWS)

AWS is Amazon's global cloud platform offering 200+ services with global infrastructure of **Regions** and **Availability Zones**.

Key Features:

- **200+ Fully Featured Services** (compute, storage, databases, networking)
- **Global Infrastructure** (Regions + AZs for high availability)
- **Pay-as-You-Go Pricing**
- **Scalable & Flexible**
- **Security & Compliance**

Core AWS Services

Compute Services

Service	Description
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Amazon EC2	Resizable virtual servers for web-scale applications
AWS Lambda	Serverless compute running code without server management

Storage & Networking

Category	Services
Storage	S3 (object storage), EBS (block storage)
Networking	VPC (virtual private cloud), Route 53 (DNS service)

Database Services

- **Amazon RDS** - Managed relational databases
- **Amazon DynamoDB** - NoSQL database with automatic scaling

AWS Pricing & Billing

Free Tier

- **12 months free access** to selected services for learning and testing

Pricing Models

Model	Description	Best For
On-Demand	Pay per hour/second	Unpredictable workloads
Reserved Instances	1-3 year commitment, up to 75% discount	Stable workloads
Spot Instances	Surplus capacity at reduced rates	Fault-tolerant jobs

Cost Management Tools

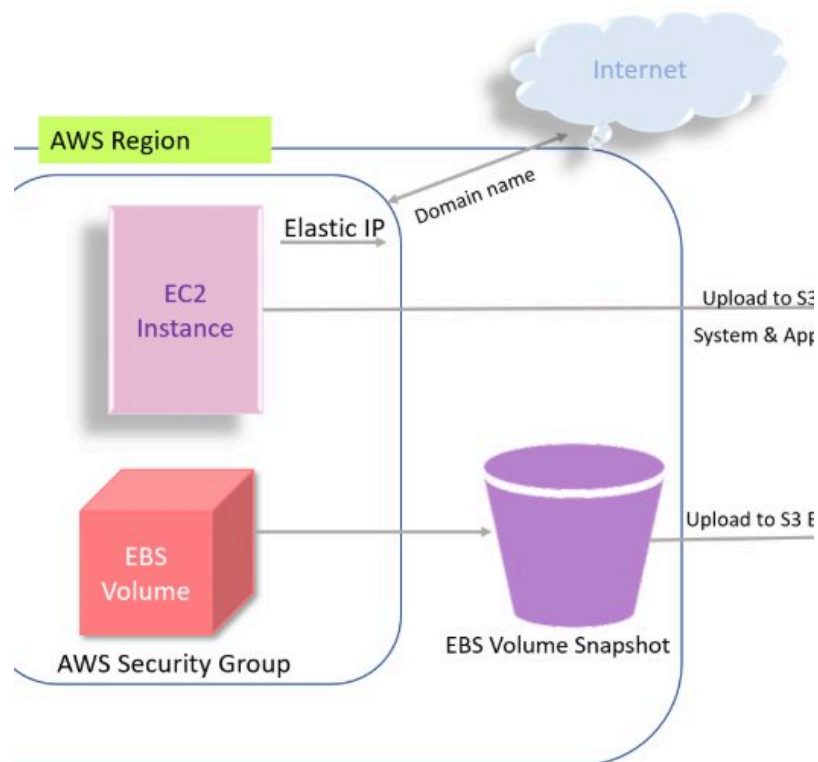
- **AWS Budgets** - Set spend limits and alerts
- **AWS Cost Explorer** - Visualize cost trends

AWS Architecture & Best Practices

Well-Architected Framework

Built on **five pillars**:

Pillar	Focus
Operational Excellence	Running and improving systems
Security	Protecting data and applications
Reliability	Ensuring correct function and recovery
Performance Efficiency	Efficient resource usage
Cost Optimization	Eliminating unnecessary costs



Scalability Tools

- **Auto Scaling** - Automatically adjusts compute capacity
- **Elastic Load Balancing (ELB)** - Distributes traffic across targets

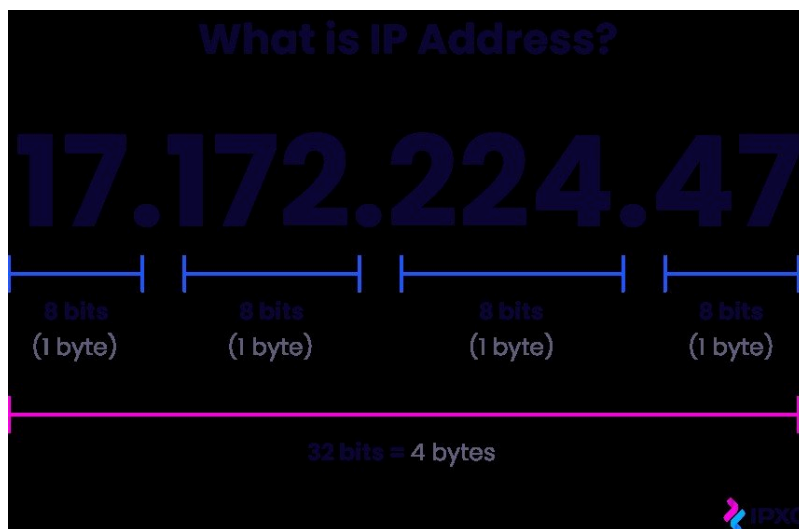


Section 2: IP Addressing Fundamentals 12/34

This section examines IP addressing concepts including binary structure, classes, and subnetting from a 14-page PDF document ([source](#))

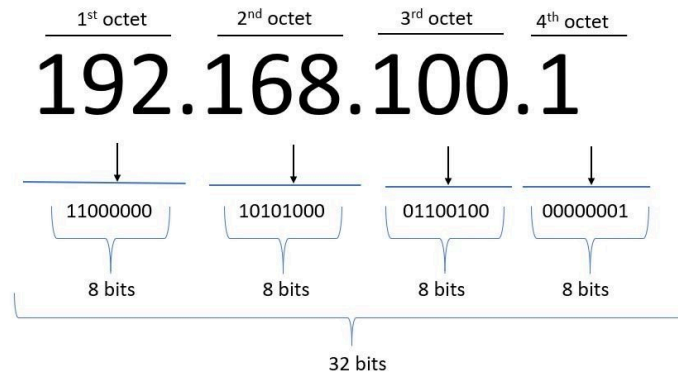
IP Address Structure

An **IP address** is a **32-bit quantity** visualized as four groups of **8 bits (octets)**.



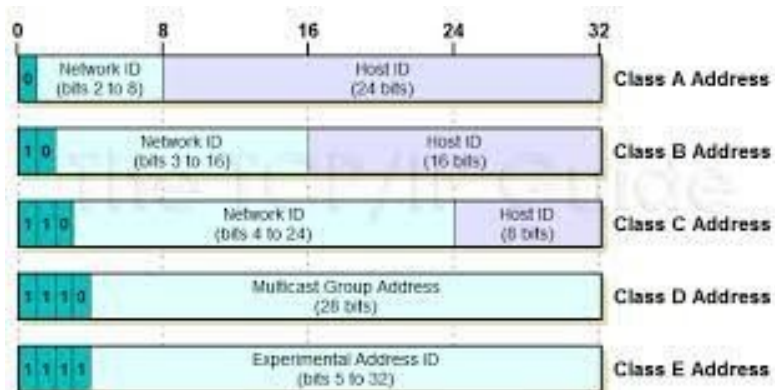
IPv4 notation: Dot-decimal (e.g., 192.168.100.1)

IPv4 Address 4-Octet



IP Address Classes

Class	IP Range	Default Subnet Mask
A	0.0.0.0 – 127.255.255.255	255.0.0.0
B	128.0.0.0 – 191.255.255.255	255.255.0.0
C	192.0.0.0 – 223.255.255.255	255.255.255.0
D	224.0.0.0 – 239.255.255.255	-
E	240.0.0.0 – 255.255.255.255	-



Subnet Masks & Subnetting

A **subnet mask** determines network ID vs. host ID portions:

Custom Subnet Mask Examples:

- 255.192.0.0 = 11111111.11000000.00000000.00000000
- 255.255.128.0 = 11111111.11111111.10000000.00000000

IP Address Types

Type	Host-portion pattern	Description
Network Address	All 0 bits	Identifies the whole network
Broadcast Address	All 1 bits	Targets all hosts in subnet
Host Address	Mixed 0 and 1 bits	Represents individual device

LAN Components

- **Switch/Hub** - Layer-2 device
- **LAN Cable** - Twisted-pair Ethernet
- **Hosts/End Devices** - Computers, printers with IP addresses

Broadcast Domains

A **broadcast domain** equals exactly one IP network address. All hosts share the same network portion.

OSI vs. TCP/IP Model

OSI Layer	TCP/IP Layer	Data Unit
Transport	Transport	Segment
Network	Internet	Packet
Data Link	Network Access	Frame
Physical	Physical	Bits

Data encapsulation: **Segment → Packet → Frame → Bits**

Section 3: IPv4 Addressing & Subnetting

This section explores IPv4 addressing structure, subnetting techniques, and network segmentation from a 44-page PDF document ([source](#))

IPv4 Address Structure

An **IPv4 address** is a **32-bit hierarchical identifier** with network and host portions determined by the **subnet mask**.

Key Concepts:

- **Subnet Mask:** Indicates network bits (1s) and host bits (0s)
- **Prefix Length:** Count of contiguous 1s (written as /n)
- **ANDing:** Logical AND operation to extract network address

IPv4 Address Types

Type	Definition	Example
Unicast	Packet to one destination	PC → Printer
Broadcast	Packet to all hosts on network	Network-wide announcement
Multicast	Packet to multicast group	PC → Group 224.10.10.5

Public vs. Private Addresses

Private IPv4 Ranges (RFC 1918):

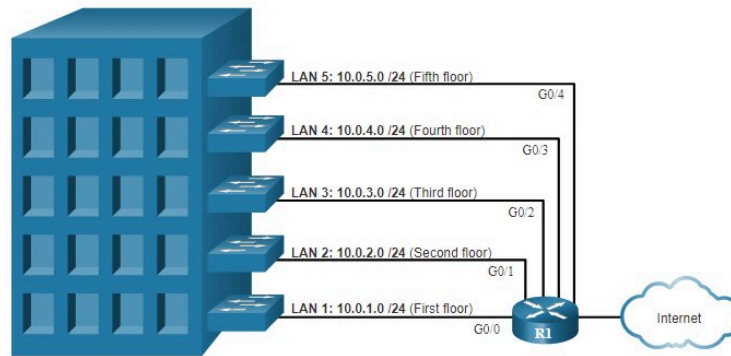
CIDR	Range
10.0.0.0/8	10.0.0.0 – 10.255.255.255
172.16.0.0/12	172.16.0.0 – 172.31.255.255
192.168.0.0/16	192.168.0.0 – 192.168.255.255

Network Segmentation

Benefits of subnetting:

- **Performance:** Reduces broadcast traffic
- **Security:** Enables ACLs per subnet

- **Management:** Logical grouping by function



Subnetting Examples

/24 Subnetting from 10.0.0.0/8

Subnet	Host Range	Broadcast
10.0.0.0/24	10.0.0.1 – 10.0.0.254	10.0.0.255
10.0.1.0/24	10.0.1.1 – 10.0.1.254	10.0.1.255

Subnetting Within /24

Prefix	Subnet Mask	Hosts per Subnet
/25	255.255.255.128	126
/26	255.255.255.192	62
/27	255.255.255.224	30
/28	255.255.255.240	14
/29	255.255.255.248	6
/30	255.255.255.252	2

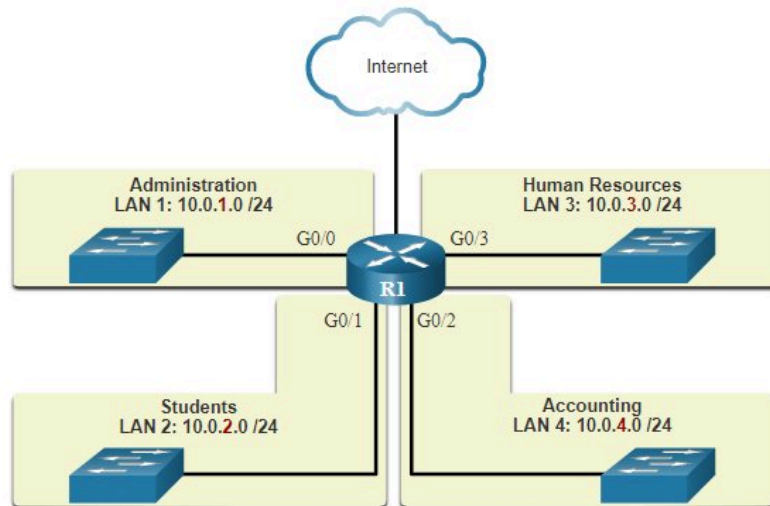
Variable Length Subnet Mask (VLSM)

VLSM allows different subnet sizes within the same network, eliminating address waste.

Design Process:

1. Identify largest host demand subnet

2. Assign smallest satisfying mask
3. Continue to next-largest requirement
4. Accommodate smallest subnet last



Section 4: Cisco Device Configuration Essentials

This section covers basic Cisco switch and router configuration, security practices, and verification techniques from a 64-page PDF document ([source](#))

Module Overview

This module teaches **Cisco device configuration using security best practices** with hands-on activities, simulations, and labs.

Switch Boot Process

Five-Step Sequence:

1. **POST** - Power-On Self-Test
2. **Boot Loader Load** - From ROM
3. **CPU Register Init** - Memory mapping
4. **Flash File System Init** - Prepares for IOS
5. **IOS Load** - Loads IOS image into RAM

Switch LED Indicators

LED	Function
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SYST	System health
STAT	Port link status
DUPLX	Duplex mode
SPEED	Speed indication
PoE	Power-over-Ethernet status

Switch Virtual Interface (SVI) Configuration

SVI provides Layer-3 IP address for remote management:

```
S1(config)# interface vlan 99
S1(config-if)# ip address 172.17.99.11 255.255.255.0
S1(config-if)# no shutdown
```

Port Configuration

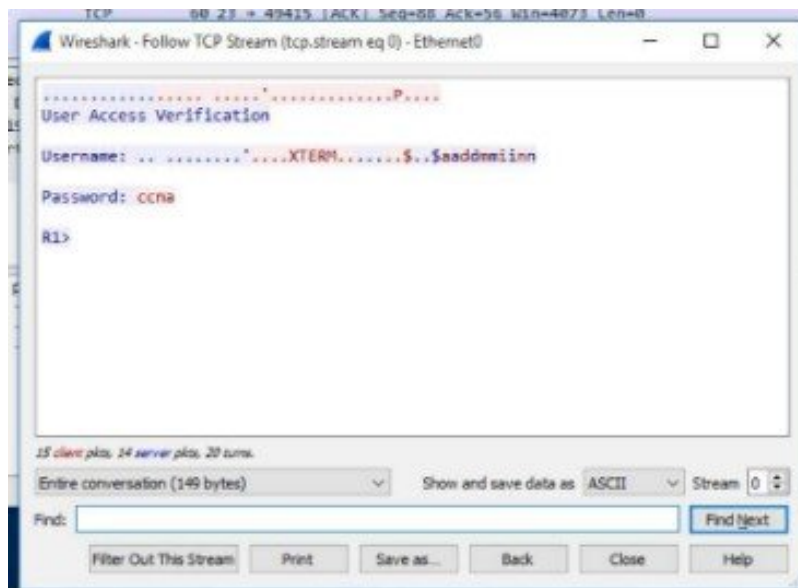
Key Settings:

- **Duplex:** full | half | auto
- **Speed:** 10 | 100 | 1000 | auto
- **Auto-MDIX:** Automatically detects cable type

Secure Remote Access

Telnet vs SSH

Protocol	Port	Security
Telnet	23	Plaintext (insecure)
SSH	22	Encrypted (secure)



SSH Configuration Steps:

1. Verify SSH capability: show ip ssh
2. Set hostname: hostname S1
3. Configure domain: ip domain-name example.com
4. Generate RSA keys: crypto key generate rsa
5. Create user: username admin secret ccna
6. Configure VTY: transport input ssh

Router Configuration Essentials

Basic Requirements:

- Hostname and passwords
- Interface IP addresses
- no shutdown command
- Banner configuration
- Configuration save

Interface Configuration:

```
Router(config)# interface gigabitethernet0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
```

Loopback Interfaces:

- Logical interfaces always up
- Useful for testing and routing protocols
- Configuration: interface loopback 0

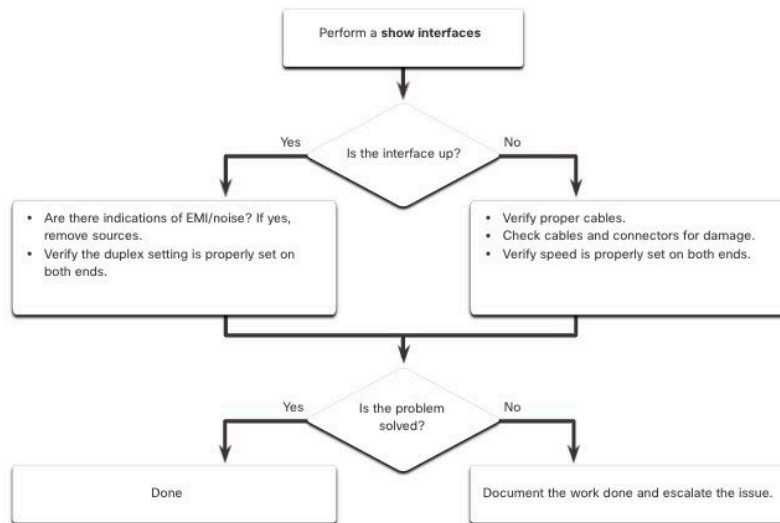
Verification Commands

Command	Purpose
show ip interface brief	Interface status
show running-config	Current configuration
show version	IOS and hardware info
show ip route	Routing table
show mac address-table	MAC addresses

```

S1# show interfaces fastEthernet 0/18
FastEthernet0/18 is up, line protocol is up (connected)
  Hardware is Fast Ethernet, address is 0025.83e6.9092 (bia 0025.83e6.9092)
  MTU 1500 bytes, BW 100000 Kbit/sec, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full-duplex, 100Mb/s, media type is 10/100BaseTX
  input flow-control is off, output flow-control is unsupported
  ARP type: ARPA, ARP Timeout 04:00:00
  Last input never, output 00:00:01, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue: 0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    2295197 packets input, 305539992 bytes, 0 no buffer
    Received 1925500 broadcasts (74 multicasts)
      0 runs, 0 giants, 0 throttles
      3 input errors, 3 CRC, 0 frame, 0 overrun, 0 ignored
      0 watchdog, 74 multicast, 0 pause input
      0 input packets with dribble condition detected
    3594664 packets output, 436549843 bytes, 0 underruns
      8 output errors, 1790 collisions, 10 interface resets
      0 unknown protocol drops
      0 babbles, 235 late collision, 0 deferred

```



```
Login as: admin
Using keyboard-interactive
Authentication.
Password:
S1> enable
Password:
S1#
```

Putting it All Together

Unified Network Foundations

- **IP addressing** underpins every communication path, whether on-premises or in the cloud.

An IPv4 address is a 32-bit identifier split into network and host portions.

- The **classful** ranges (A,B,C) give way to **CIDR** notation (/n) that defines the subnet mask and the size of each broadcast domain.
- **Subnet masks** (e.g., 255.255.255.192) act as binary filters, separating network bits (1) from host bits (0).

- **Broadcast domains** are synonymous with a single IP network; reducing their size through **subnetting** or **VLANs** improves performance and security.

Concept	Role in the network
Network address	First address in a subnet (host bits = 0)
Broadcast address	Last address in a subnet (host bits = 1)
Host address	Any address between network and broadcast
VLSM	Tailors subnet size to actual host needs, eliminating waste

Cloud & On-Prem Integration

- **Cloud computing** delivers compute, storage, and services over the Internet, providing **elastic scalability** and **pay-as-you-go** economics.

Cloud services are accessed remotely, removing the need for local hardware ownership.

- **AWS** organizes its infrastructure into **Regions** and **Availability Zones (AZs)**, guaranteeing high availability and fault tolerance.
- Core AWS services map directly to traditional network components:

AWS Service	Traditional equivalent
EC2	Virtual servers (compute)
S3 / EBS	Object & block storage
VPC	Isolated virtual network
Route 53	DNS routing
RDS / DynamoDB	Managed relational & NoSQL databases

- When extending a **private LAN** to the cloud, the same **subnetting principles** apply: allocate a CIDR block (e.g., 10.0.0.0/16) and carve out sub-nets for each AZ, mirroring on-prem VLAN segmentation.

Security & Access

- **SSH** supersedes Telnet by encrypting both credentials and traffic; it requires a **k9-enabled IOS** image and RSA key generation.

```
crypto key generate rsa
username admin secret
line vty 0 4
  transport input ssh
  login local
```

- **Switch Virtual Interfaces (SVI)** give a Layer-3 management address without occupying a physical port. Assign the SVI to a **dedicated management VLAN** (e.g., VLAN 99) and configure a default gateway for off-subnet reachability.
- **Port security** is reinforced by matching **speed/duplex** settings and enabling **auto-MDIX** to avoid cable-type mismatches.

Security measure	Typical command
Enable SSH (v2)	ip ssh version 2
Set RSA key size	crypto key generate rsa modulus 2048
Define management VLAN	interface vlan 99 → ip address 172.17.99.11 255.255.255.0
Restrict VTY access	line vty 0 4 → login local

Management, Verification & Cost Optimisation

- **Show commands** (show ip interface brief, show running-config, show mac address-table) provide real-time visibility of interface status, VLAN membership, and routing information.
- **AWS Cost tools** (Budgets, Cost Explorer) parallel on-prem **resource monitoring** (show version, show interfaces) by highlighting utilization trends and alerting on overruns.
- **Auto Scaling** and **Elastic Load Balancing** in AWS embody the same principles as **router-based load distribution** and **switch port aggregation**: keep traffic flowing while using the minimum necessary resources.

Verification tool	On-prem analogue
show ip route (C/L entries)	Routing table inspection
AWS Budgets	Network device alert thresholds
Cost Explorer	Interface statistics (show interfaces)
Auto Scaling	Switch port speed/duplex auto-negotiation

Practical Workflow Synthesis

1. Design the address plan

- Choose a private CIDR block (e.g., 10.0.0.0/8).
- Apply **VLSM** to allocate sub-nets that meet host requirements (e.g., /26 for 60 hosts, /30 for point-to-point links).

2. Implement on-prem segmentation

- Create VLANs, assign SVIs, and configure **default gateways**.
- Verify with show ip interface brief and show vlan.

3. Extend to the cloud

- Provision matching VPC sub-nets in each AWS AZ.
- Deploy EC2 instances and attach them to the appropriate subnet.

4. Secure remote management

- Enable SSH on both switches and EC2 instances.
- Use strong local user credentials and RSA keys.

5. Monitor and optimise

- Track on-prem interface utilization (show interfaces) alongside AWS Cost Explorer reports.
- Adjust subnet sizes or scale EC2 capacity based on observed trends.

Key takeaways

- **Consistent addressing** (CIDR, VLSM) bridges physical LANs, VLANs, and cloud VPCs.
 - **Security** hinges on encrypted management (SSH), proper VLAN isolation, and least-privilege access.
 - **Automation & scaling**—whether through Auto Scaling groups in AWS or dynamic port configurations on Cisco gear—ensure resources match demand without waste.
-